Phys. 2110 HW 13: 13.9, 11, 12, 26, 44, 53, 82, 84

13.9 (a) $T = 2\pi\sqrt{\frac{0.500\ kg}{140\ N/m}} = 0.375\ s$

(b) $f = \frac{1}{T} = 2.66\ Hz$ (c) $\omega = 2\pi f = 16.7\ rad/s$

13.11 (a) From Eq. (13.19), $A = \left|\frac{V_0}{\omega}\right| = \left|\frac{V_0}{\sqrt{K/m}}\right| = 0.98\ m$

(b) Since $x(0) = 0$, Eq. (13.14) $x_0 = A\cos\phi$ requires $\phi = \pm\frac{\pi}{2}$.
Since the block is initially moving in the negative direction, $V_{0x} < 0$,
Eq. (13.17) $V_{0x} = -\omega A \sin\phi$ requires $\sin\phi > 0$, so $\phi = \frac{\pi}{2}$.

(c) $x = (0.98m)\cos(\omega t + \frac{\pi}{2}) = (-0.98m)\sin((12.2\ rad/s)t)$

13.12 (a) Eq. (13.19): $A = \sqrt{x_0^2 + V_{0x}^2/\omega^2} = \sqrt{x_0^2 + mV_{0x}^2/k}$

$A = \sqrt{(0.200m)^2 + (2.00\ kg)(-4.00m/s)^2/(300\ N/m)} = 0.383\ m$

(b) Eq. (13.18): $\phi = \arctan(-V_{0x}/\omega x_0)$

$\omega = \sqrt{\frac{k}{m}} = 12.25\ rad/s$

$\phi = \arctan\left(\frac{4.00\ m/s}{(12.25\ rad/s)(0.200m)}\right) = \arctan(1.633) = 58^\circ\ (1.02\ rad)$

(c) $x = A\cos(\omega t + \phi)$ gives $x = (0.383m)\cos((12.2\ rad/s)t + 1.02\ rad)$

13.26 (a) $E = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$

$E = \frac{1}{2}(0.150\ kg)(0.300\ m/s)^2 + \frac{1}{2}(300\ N/m)(0.012\ m)^2 = 0.0284\ J$

(b) $E = \frac{1}{2}kA^2$ so $A = \sqrt{\frac{2E}{k}} = \sqrt{\frac{2(0.0284J)}{300\ N/m}} = 0.014\ m$

(c) $E = \frac{1}{2}mV_{max}^2$ so $V_{max} = \sqrt{\frac{2E}{m}} = \sqrt{\frac{2(0.0284J)}{0.150\ kg}} = 0.615\ m/s$

13.44 The period $T = 2\pi\sqrt{\frac{L}{g}}$

$\frac{T_M}{T_E} = 2\pi\sqrt{\frac{L}{g_M}} \Big/ 2\pi\sqrt{\frac{L}{g_E}} = \sqrt{\frac{g_E}{g_M}}$

$T_M = \sqrt{\frac{g_E}{g_M}}\ T_E = \sqrt{\frac{9.80m/s^2}{3.71m/s^2}}\ (1.60s) = 2.60s$

13.53 Pendulum A : $T_A = 2\pi\sqrt{\frac{L}{g}}$

Pendulum B : $I = I_{bar} + I_{ball} = \frac{1}{3}\left(\frac{m}{2}\right)L^2 + \left(\frac{m}{2}\right)L^2 = \frac{2}{3}mL^2$

$$T_B = 2\pi\sqrt{\frac{I}{mgd}} = 2\pi\sqrt{\frac{\frac{2}{3}mL^2}{mg\left(\frac{3}{4}L\right)}} = \sqrt{\frac{8}{9}}\left(2\pi\sqrt{\frac{L}{g}}\right) = \sqrt{\frac{8}{9}}\,T_A.$$

The period is longer for pendulum A.

13.82 Example 12.10 derives $F_r(r) = -\frac{GM_E m}{R_E^3} r$

$a_r = \frac{F_r}{m} = -\frac{GM_E}{R_E^3} r$ is in the form of Eq. (13.8), with x replaced by r, so the motion is simple harmonic.

$F_r(r) = -kr$ to determine k. $\quad k = \frac{GM_E m}{R_E^3}$

$$\omega^2 = \frac{k}{m} = \frac{GM_E}{R_E^3} = \frac{g}{R_E}$$

The period is then $T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{R_E}{g}} = 2\pi\sqrt{\frac{6.38\times10^6\,m}{9.80\,m/s^2}} = 5070\,s = 84.5\,min$

13.84 (a) $U = -\int_0^x F_x\,dx = c\int_0^x x^3\,dx = \frac{c}{4}x^4$

(b) From conservation of energy, $\frac{1}{2}mv_x^2 = \frac{c}{4}(A^4 - x^4)$

$v_x = \frac{dx}{dt}$, so $\frac{dx}{\sqrt{A^4-x^4}} = \sqrt{\frac{c}{2m}}\,dt \qquad \int_0^A \frac{dx}{\sqrt{A^4-x^4}} = \int_0^{\frac{T}{4}}\sqrt{\frac{c}{2m}}\,dt = \frac{T}{4}\sqrt{\frac{c}{2m}}$

change of variable $u = \frac{x}{A}$

$$\frac{1}{A}\int_0^1 \frac{du}{\sqrt{1-u^4}} = \frac{1.31}{A} = \frac{T}{4}\sqrt{\frac{c}{2m}} \quad \text{so} \quad T = \frac{7.41}{A}\sqrt{\frac{m}{c}}$$

(c) The period does depend on amplitude, and the motion is not simple harmonic.